Modelling mass TB and TBI screening in South Tarawa (Kiribati) — Methods appendix

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Contents

1 Overview of the Model and Analysis

We developed a deterministic, age-structured compartmental model of *Mycobacterium tuberculosis* (M.tb) transmission, infection, disease, detection, treatment, and prevention. The model uses discrete age groups and continuous time, and is formulated as a system of ordinary differential equations (ODEs). All compartments are stratified by age group and by screening reachability, so that each model state is indexed by both dimensions.

Compartment notation (per age group and reachability stratum). We denote the following compartments:

- **S**: M.tb-*naïve* (never infected).
- **E**: Incipient infection (post-infection with risk of disease progression).
- **C**: Contained infection (immune control without sterilisation - no immediate disease progression risk).
- **X**: Cleared infection (post infection clearance - spontaneous or TPT-induced).
- $\mathbf{D_{SL}}$: Active TB, Subclinical, Low infectiousness.
- $\mathbf{D_{CL}}$: Active TB, Clinical, Low infectiousness.
- $\mathbf{D_{SH}}$: Active TB, Subclinical, High infectiousness.
- $\mathbf{D_{CH}}$: Active TB, Clinical, High infectiousness.
- **T**: On TB treatment.
- **R**: Recovered (post self-recovery or treatment).

We use the term "M.tb naïve" rather than the commonly used "Susceptible" to describe individuals who have never been infected with M.tb, because several non-naïve compartments remain (partially) susceptible to reinfection.

Model structure. Figure 1 shows the model compartments and the transitions between them.

Model overview. Non-disease *M. tuberculosis* infection is modelled through two stages: incipient (E) and contained (C). The contained stage represents a state in which the infection is controlled by the immune system, with no immediate risk of progression to active disease. Breakdown of containment reflects loss of immune control, returning individuals to the incipient stage (E) with a renewed risk of disease progression. Containment can also result in complete clearance of infection, represented by state X.

Progression from E results in subclinical active disease (more or less infectious) that may gain/lose infectiousness and progress/regress between clinical and subclinical states. More details about the disease categorisation and transitions are provided in Section 4. Only *clinical* forms of disease are detected through passive case finding; while active screening may potentially detect any form of TB. Screening programmes add time-limited detection flows for both TB disease and non-disease M. tb infection states (Section 7).

Age-specific demography and mixing are implemented explicitly to capture the effects of population structure and contact patterns on transmission dynamics (Sections 3–5.2).

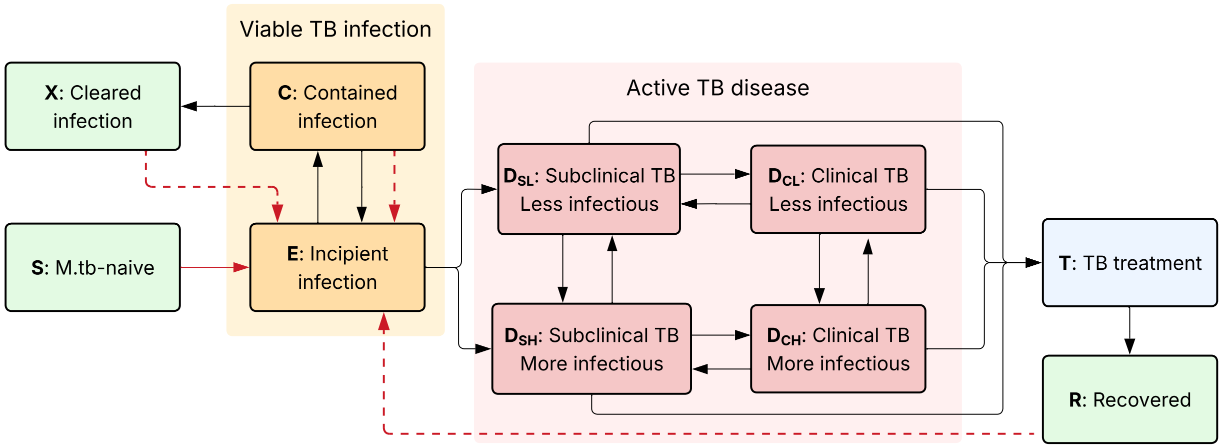

Figure 1: **Schematic of model compartments and transitions.** Transmission pathways are shown in red, with dashed red arrows indicating reinfection. For clarity, age and screening-reachability stratification, births, natural mortality, spontaneous recovery from subclinical TB, and TB-related mortality from clinical TB are not shown.

The model is additionally stratified by screening reachability, distinguishing individuals who can participate in active screening from those who are hard to reach. The latter receive no active screening and are assumed to have greater susceptibility to infection and poorer access to passive case detection (Section 6).

Simulation period and initial conditions. The model is simulated from 1850 to 2035. The initial population size is set to the earliest available WPP estimate for the modelled population (1950), and no population growth is applied before that first data year, so the population size remains constant over the initial period and follows the observed trajectory thereafter (Section 3). The epidemic is seeded at the start of the simulation with 100 individuals in the clinical, high-infectiousness disease state (D_{CH}), with the remainder of the population placed in the M.tb-naïve compartment (S). The long period preceding the first calibration data allows the age distribution and the TB epidemic to reach an endemic equilibrium well before the period informed by observations, so that results are insensitive to the size and composition of the initial seed.

Parameter treatment in calibration. Parameters with specified prior distributions are estimated in the Bayesian calibration while those without priors are fixed. Estimated classes include core transmission (baseline transmission rate and population-scaling exponent), social mixing (background mixing level, age assortativity spread, and parent–child mixing strength), selected early-infection and disease-transition rates (clearance, immunity breakdown, clinical progression, gain of infectiousness), the passive-detection scale-up profile, and a subset of diagnostic sensitivities.

Model use in analysis. The model is used in two stages: (i) to reconstruct the historical epidemic trajectory by calibrating uncertain parameters to epidemiological and intervention data; and (ii) to project forward under alternative intervention scenarios (screening and preventive treatment). Parameter estimation is performed in a Bayesian framework with Differential Evolution Metropolis algorithm (DEMetropolisZ); posterior samples are propagated to quantify uncertainty in projected outcomes.

2 Software and Code

The model is implemented in Python using `summer2`, enabling symbolic specification and efficient numerical ODE generation. Computations are accelerated via `jax` just-in-time compilation, which lowers the ODE system to optimised machine code after first execution; subsequent simulations require ∼200 ms each, enabling large-scale Bayesian calibration.

All model code, calibration routines, analysis scripts, and interactive notebooks are available at `https://github.com/monash-emu/kiribati_tb_modelling`.

The repository includes instructions for installation and reproduction of results. The project environment is managed using `pixi` for reproducible dependency management and straightforward installation.

3 Demographics

Each age and disease state in the model is subject to the appropriate all-cause and TB-specific mortality rates. Non-TB deaths are calculated using time-variant and age-specific mortality rates, and births are injected into the youngest M.tb-naïve compartment to reproduce the observed population size over time. This approach allows the model to match observed population trajectories while maintaining internal demographic dynamics. Age-specific mortality rates and population size over time are taken from the United Nations World Population Prospects (WPP) data. We implement ageing as first-order transitions between adjacent age groups, with rates calculated as the inverse of the age group's span and applied to all age strata except the oldest one.

4 Natural History of Infection and Disease

We represent early TB infection dynamics using two non-disease states: an *incipient* state (E) and a *contained* state (C). Individuals newly infected with *M. tb* enter E, where infection is unstable and at risk of progression. From E, they may either progress to subclinical active TB or undergo immune-mediated containment, moving to C. Contained infection may subsequently clear entirely to X (uninfected/cleared) or experience immunological failure, returning to the incipient state E. Active TB is stratified by both clinical presentation (subclinical vs. clinical) and infectiousness (low vs. high), yielding four disease compartments: $D_{SL}, D_{CL}, D_{SH}, D_{CH}$. Individuals with subclinical disease may progress to clinical disease, and clinical disease may regress to subclinical; similarly, infectiousness may increase (low $\rightarrow$ high) or decrease (high $\rightarrow$ low). However, transitions are restricted to a single dimension at a time. Individuals may change clinical status (subclinical $\leftrightarrow$ clinical) or infectiousness (low $\leftrightarrow$ high), but simultaneous transitions in both dimensions are not allowed. Self-recovery is allowed only from subclinical states, with recovered individuals entering R. In contrast, TB-induced mortality rates are only applied to the clinical states. Treatment is available for all active states, with successful treatment leading to recovery (R) and unsuccessful treatment resulting in relapse to subclinical low infectiousness (D_{SL}) or death.

Early infection. Progression from E is age-specific (three categories: < 5, 5–14, 15+) and partitioned into low vs high infectiousness at first presentation. In the absence of data informing this split, progression from incipient infection is divided equally between the less-infectious and the more-infectious subclinical states ($p = 0.5$). This assumption is not restrictive, because the subsequent within-disease transition rates are estimated during calibration, so that the resulting distribution of prevalent TB across the four disease states reproduces the observed data regardless of the initial split. Containment rates are also age-specific. Clearance and immune breakdown are age-invariant.

Clinical and infectiousness transitions. Movements between subclinical and clinical states, and between low and high infectiousness, govern within-disease dynamics. The cross-sectional observations available from the completed PEARL screening phase constrain the relative sizes of the four disease compartments, but not the speed at which individuals move between them: the same distribution of prevalent TB across states can be reproduced by slow or by rapid transitions in both directions.

To resolve this identifiability problem, the two regression rates (clinical → subclinical and high → low infectiousness) are fixed at a common value, while the two corresponding progression rates (subclinical → clinical and low → high infectiousness) are estimated during calibration. In the base case the common regression rate is set to 1 per year, corresponding to an average of one year spent in a given state before regression. Under this constraint, the estimated progression rates adjust so that the model reproduces the observed distribution of prevalent TB across the four states. The priors placed on the two estimated rates were chosen wide enough that the posteriors are determined by the PEARL observations rather than by the prior bounds.

Because this common value is an assumption rather than an estimate, it is varied in sensitivity analysis across 0.5, 1, 2 and 3 per year, with the model recalibrated at each value. This yields configurations that all reproduce the observed cross-sectional distribution of disease states while differing markedly in the overall speed of movement across the TB spectrum.

Infectiousness weighting. Let σ_S denote the relative infectiousness of subclinical vs clinical disease, and σ_L the relative infectiousness of low vs high. Define the active-disease index set

$$\mathcal{K} = \{SL, CL, SH, CH\},$$

with relative infectiousness weights w_k given by $w_{CH} = 1$, $w_{CL} = \sigma_L$, $w_{SH} = \sigma_S$, $w_{SL} = \sigma_S \sigma_L$. Children under 15 years contribute zero infectiousness, implemented by setting $w_k = 0$ in those age groups.

4.1 Generalised Transmission and Infectiousness Weights

We interpolate between frequency- and density-dependent transmission using the exponent $\kappa \in [0, 1]$ applied to the contacted population size. Because mixing between reachability strata is homogeneous (Section 6), the force of infection depends on age group only. For age group i it is

$$\lambda_i(t) = \beta_0 \, N_{\text{ref}}^{\kappa} \sum_j \tilde{C}_{ij}(t) \, \frac{I_j^{\text{eff}}(t)}{N_j(t)^{\kappa}},$$

where $\tilde{C}(t)$ is the spectral-radius-normalised age-mixing matrix (Section 5.2). Here N_{ref} denotes the total population in year 2020 and is included purely as a scaling constant for more efficient calibration. This rescaling stabilises the magnitude of the transmission coefficient β_0 across different values of κ, preventing its calibrated value from varying by orders of magnitude as the dependence on population size changes. The effective infectious pool and the population size of age group j are both aggregated over the two reachability strata $s \in \{\mathrm{r}, \mathrm{u}\}$:

$$I_j^{\text{eff}}(t) = \sum_s \sum_{k \in \mathcal{K}} w_k \, D_{k,j,s}(t), \qquad N_j(t) = \sum_s N_{j,s}(t).$$

Infectiousness weights w_k do not differ by reachability stratum; only susceptibility does, through the multiplier θ_s^{sus} introduced in Section 6.

4.2 Reinfection Pathways

Reinfection is permitted from C, X, and R with relative susceptibilities ρ_C (contained) and $\rho_X = \rho_R$ (cleared and recovered), and a child susceptibility modifier σ^{child} applied when infection arises from S in $< 15\,\mathrm{y}$ groups. This adjustment captures the effect of BCG vaccination, which is typically administered in infancy and provides protection against infection for several years.

5 Age Stratification and Mixing

5.1 Ageing and Age-Specific Adjustments

Age groups. The population is stratified into $G = 8$ age groups with lower bounds

$$\mathcal{A} = \{0, 3, 5, 10, 15, 18, 40, 65\}.$$

The final group is open-ended (65+).

The breaks were chosen so that every age-dependent process and observation used in the analysis aligns exactly with a group boundary, avoiding the need to apportion parameters or targets within groups:

- **3 years** is the minimum age of eligibility for screening under the PEARL algorithm, below which no active screening or preventive treatment is offered.
- **5 and 15 years** delimit the three age categories used for the early-infection parameters (< 5, 5–14, 15+), and 15 years additionally separates the age groups in which susceptibility to infection is reduced to reflect BCG protection and infectiousness is set to zero.
- **10 years** separates children receiving symptom screening only from older individuals additionally receiving chest X-ray and, in the PEARL algorithm, frontline Xpert.
- **18 years** allows the model to reproduce the earlier estimate of TST positivity among adults aged 18 years and over, which is used as a calibration target (Section 9.1).
- **40 and 65 years** subdivide the adult population to capture differences in background mortality and in contact patterns across adult ages.

Ageing and adjustments. Ageing is implemented as first-order transitions between adjacent age groups. Several age-specific adjustments and interventions are applied in the model (see previous sections for details):

- Age-specific early infection rates using three categories: < 5 years, 5–14 years, and 15+ years.
- Reduced susceptibility to infection from S in individuals aged < 15 years to reflect BCG effect.
- Infectiousness set to zero for individuals aged < 15 years.
- Age-specific background mortality.
- Age-specific screening interventions.

5.2 Age-Mixing Matrix

Notation and data sources. Let $i, j \in \{1, \dots, G\}$ index age groups and $[a_i, b_i]$ denote the single ages in group i. Calendar time is t (years). We use UN World Population Prospects data to inform age-specific population sizes (1950–2100) and age-specific fertility rates (1950–2023). For modelled years outside data ranges, quantities are clamped to the nearest available year.

Overview. At time t, we construct an age-specific contact matrix

$$\mathbf{C}(t) = \big(C_{ij}(t)\big)_{i,j=1}^{G},$$

where $C_{ij}(t)$ is the per-capita rate at which age group i contacts age group j. Construction proceeds by: (1) building a symmetric per-pair intensity matrix $\mathbf{S}(t)$; (2) scaling columns by the contacted group size; (3) normalising by the spectral radius. We combine three components: uniform background mixing, assortative age mixing, and parent–child mixing. Three scalar parameters govern these components (background level, assortativity spread, and parent–child mixing strength) and are estimated.

Population weights within age groups. To allow non-uniform single-age distributions within groups, let $N(a, t)$ be the population at age a and time t. For group i,

$$w_i(a,t) = \frac{N(a,t)}{\sum_{a' \in [a_i,b_i]} N(a',t)}, \qquad \sum_{a \in [a_i,b_i]} w_i(a,t) = 1.$$

Symmetric per-pair intensity. For each i, j,

$$S_{ij}(t) = b + S_{ij}^{\text{assort}}(t) + \kappa_{\text{pc}}\, S_{ij}^{\text{pc}}(t), \quad S_{ij}(t) = S_{ji}(t),$$

where b is the background level (`bg_mixing`), S^{assort} is an exponential kernel with spread ℓ (`a_spread`), and S^{pc} encodes parent–child contacts from fertility patterns, with strength κ_{pc} (`pc_strength`). Smaller values of ℓ correspond to more strongly assortative mixing by age.

Assortative age mixing.

$$S_{ij}^{\text{assort}}(t) = \sum_{a \in [a_i,b_i]} \sum_{c \in [a_j,b_j]} w_i(a,t)\, w_j(c,t)\, \frac{1}{\ell} \exp\!\left(-\frac{|a-c|}{\ell}\right).$$

Parent–child mixing. Let $p_{\text{pc}}(\Delta, y)$ be the probability of a parent–child age gap Δ for a child born in year y. With $a_{\text{child}} = \min(a, c)$ and birth year $t - a_{\text{child}}$,

$$S_{ij}^{\text{pc}}(t) = \sum_{a \in [a_i,b_i]} \sum_{c \in [a_j,b_j]} w_i(a,t)\, w_j(c,t)\, p_{\text{pc}}\big(|a-c|,\ t - a_{\text{child}}\big).$$

Population scaling and asymmetry. The per-pair matrix becomes an asymmetric WAIFW matrix by column scaling with the contacted group size:

$$C_{ij}(t) = S_{ij}(t)\, N_j(t).$$

Normalisation. Let $\rho(t)$ be the spectral radius of $\mathbf{C}(t)$. We use the normalised matrix

$$\tilde{\mathbf{C}}(t) = \frac{\mathbf{C}(t)}{\rho(t)}.$$

6 Stratification by Screening Reachability

Population-wide screening never reaches the entire population, and the individuals who are not reached may differ epidemiologically from those who participate. To capture this, the model is stratified into two mutually exclusive strata: a *screening-reachable* stratum, comprising individuals who can participate in active screening, and a *hard-to-reach* stratum, comprising individuals who do not. This stratification is applied to all compartments and is crossed with the age stratification, so that each model state is indexed by both age group i and reachability stratum $s \in \{\mathrm{r}, \mathrm{u}\}$.

Population split. The proportion of the population in the screening-reachable stratum is denoted π and is fixed at 0.85, based on participation observed during the completed phase of the PEARL study. The initial population is split between the two strata according to π and $1 - \pi$ within every compartment and age group.

The stratum sizes are held at these proportions over time. Because there are no transitions between the screening-reachable and hard-to-reach strata, this is achieved by (i) splitting the additional births used to reproduce population growth between the two strata according to π and $1 - \pi$, and (ii) returning all deaths (background, TB-related, and on-treatment) to the M.tb-naïve compartment of the youngest age group *within the same reachability stratum* from which they originated. Reachability is therefore a fixed, lifelong attribute in the model, and the fraction of the population that is hard to reach does not drift as the epidemic or the interventions evolve. Mixing between the screening-reachable and hard-to-reach strata is assumed to be homogeneous. In the absence of data on the contact patterns of individuals not reached by screening, this is a deliberately conservative assumption: assortative mixing within the hard-to-reach stratum would be expected to further concentrate residual transmission in that group.

Epidemiological differences between strata. Barriers to participating in screening are unlikely to occur in isolation and may coincide with living conditions or health-seeking behaviours that also affect TB risk and access to routine diagnosis. Two multiplicative adjustments are therefore applied to the hard-to-reach stratum, relative to the screening-reachable stratum:

- **Susceptibility to infection.** All infection and reinfection flows (from S, C, X and R) are multiplied by θ_{sus} (`rel_sus_unreachable`) in the hard-to-reach stratum. The base-case value is 1.5, i.e. hard-to-reach individuals are assumed to acquire infection at 1.5 times the rate of screening-reachable individuals experiencing the same force of infection. This factor is applied to the susceptibility side of transmission only; the relative infectiousness of individuals with TB does not differ by stratum.
- **Passive detection.** The time-varying passive detection rate $\delta(t)$ (Section 7) is multiplied by θ_{det} (`rel_detection_unreachable`) in the hard-to-reach stratum. The base-case value is 0.5, reflecting an assumption of reduced access to routine diagnostic services in the same population that is difficult to reach through active screening.

Both parameters are fixed rather than estimated, as the available data do not inform them; θ_{sus} is varied in sensitivity analysis.

Access to screening. All active screening flows, for both TB disease and non-disease M.tb infection, are set to zero in the hard-to-reach stratum. Screening programmes therefore act exclusively on the screening-reachable stratum. As a consequence, the maximum achievable population coverage is π: a nominal coverage of 85% of the total population corresponds to screening essentially the whole screening-reachable stratum, whereas lower nominal coverage levels represent incomplete participation among screening-reachable individuals (Section 7).

Calibration and reporting. Because the epidemiological observations collected during the completed PEARL screening phase were made among participants, the corresponding calibration targets (measured TB prevalence and TST positivity) are compared with model outputs restricted to the screening-reachable stratum. Historical notifications, which arise from passive detection in the whole population, are compared with model outputs aggregated across both strata. Projected burden outcomes are likewise reported for the entire modelled population unless otherwise stated, and the share of TB incidence arising from the hard-to-reach stratum is tracked as an additional output.

7 TB Detection, Treatment and Screening

7.1 Detection

Passive detection. Passive case finding applies *only to clinical disease*. Subclinical disease is asymptomatic by definition and is therefore assumed not to prompt care seeking, so that individuals with subclinical TB can enter treatment only after progressing to clinical disease, or through active screening.

The passive detection rate is allowed to vary over time, because diagnostic capacity and access to care in Kiribati have strengthened progressively over recent decades, and a constant rate would be difficult to reconcile with the historical notification series. Denote the time-varying passive detection profile by $\delta(t)$, which follows a smooth hyperbolic-tangent scale-up from an earlier (lower) level to a recent level:

$$\delta(t) = \delta_{\text{rec}} \left[\phi + (1-\phi) \left(\frac{\tanh\big(\sigma\,(t-t^*)\big)}{2} + \frac{1}{2} \right) \right].$$

Here, δ_{rec} is the recent detection level, $\phi \in (0,1]$ is the past-to-recent ratio, $\sigma > 0$ controls steepness, and t^* is the inflection time.

Figure 2 illustrates an example profile for $\delta_{\text{rec}} = 0.7$, $\phi = 0.5$, $\sigma = 0.15$ and $t^* = 2005$.

7.2 Screening programmes

We consider a set of time-limited screening interventions implemented over the year 2026. Each screening programme is defined by a screening algorithm, a target population (by age), and a coverage level. Three coverage levels are explored: 65%, 75% and 85%. These are expressed as proportions of the *total* target population, that is, with hard-to-reach individuals included in the denominator.

Because screening can only reach the 85% of the population classified as screening-reachable (Section 6), this represents the theoretical maximum population coverage. In practice, the scenarios labelled as “85% coverage” are implemented using 84.5% coverage, because complete coverage of the screening-reachable stratum would require an infinite screening rate under the constant-hazard formulation described below. The two lower levels represent incomplete participation among screening-reachable individuals.

For each programme, the total coverage is translated into a constant per-capita screening rate applied to the screening-reachable stratum over the intervention period. Specifically, if c denotes

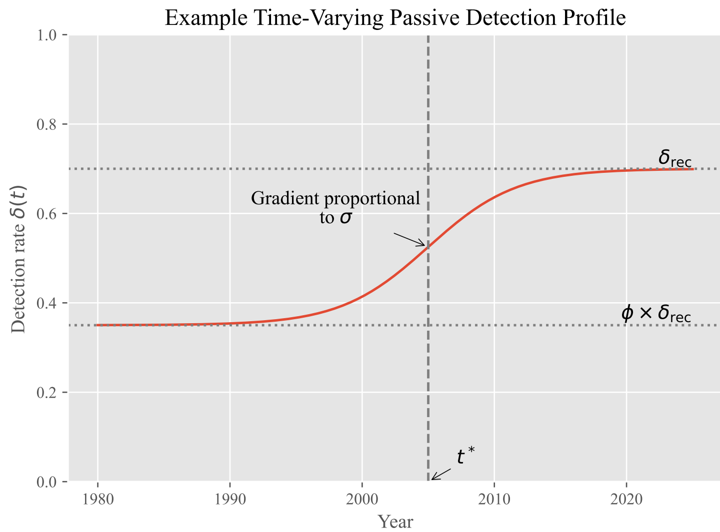

Figure 2: Example time-varying passive detection profile for clinical TB disease.

the target coverage of the total population (expressed as a proportion), π the screening-reachable fraction, and Δt the duration of the campaign, the screening rate is defined as $-\frac{\log(1-c/\pi)}{\Delta t}$.

This ensures that, under a constant hazard assumption, a proportion c/π of the eligible screening-reachable population, and hence a proportion c of the total target population, is screened over the intervention period. The screening rate is implemented as a time-varying function that is zero outside the intervention window and constant during the campaign.

Screening interventions may act on TB infection or active TB disease states. For each screening algorithm, screened individuals have a probability of being correctly identified defined by the sensitivity of the relevant algorithm for that compartment. For TB screening, positively screened individuals are moved to the treatment compartment (T). For TPT, only a fraction of the positively screened individuals transition to the cleared compartment (X) based on the probability of successful TPT completion.

Screening algorithms. Sensitivities in the model are properties of a complete *algorithm* rather than of an individual test. For a given algorithm, the sensitivity assigned to a compartment is the probability that a person occupying that state, and screened under that algorithm, is correctly identified and referred for treatment.

We consider the following approaches for the detection of TB disease, in increasing order of intensity:

- **Symptom screening (SSx)**: screening on the basis of reported symptoms alone. Clinical disease is assumed to be identified with certainty and subclinical disease not at all, so that this algorithm cannot detect subclinical TB by construction.

- **Chest X-ray (CXR)**: symptom screening combined with chest X-ray, with bacteriological confirmation of those with an abnormal result. The added value of CXR lies in the detection of subclinical disease, and the corresponding sensitivities are estimated during calibration (separately for less- and more-infectious subclinical TB), informed by the prevalence observed during the completed PEARL screening phase.

- **PEARL**: symptom screening and CXR, with Xpert MTB/RIF Ultra additionally used as a frontline test in a subset of participants. Operational constraints during implementation meant that frontline Xpert was applied to 35% of those screened, so the algorithm is represented as a mixture of the two branches:

$$s_k^{\text{PEARL}} = 0.35\, s_k^{\text{Xpert+CXR}} + 0.65\, s_k^{\text{CXR}}, \qquad k \in \mathcal{K},$$

  where $s_k^{\text{Xpert+CXR}}$ is the sensitivity of the branch in which frontline Xpert is added to symptom screening and CXR, fixed at one for all four disease states. The weights are the proportions of participants allocated to each branch, so that s_k^{PEARL} is the expected probability of detection for a randomly selected person screened under the programme as it was actually delivered, rather than under either branch in isolation.

By construction, $s_k^{\text{SSx}} \leq s_k^{\text{CXR}} \leq s_k^{\text{Xpert+CXR}}$ for every disease state k.

Screening for non-disease M.tb infection is performed using the **tuberculin skin test (TST)**, applied to the incipient and contained states with a common probability of a positive result. Individuals identified in this way are offered TPT, and only those completing it transition to the cleared state.

TST reactivity is not restricted to these two states. Individuals who have cleared infection or recovered from disease may also react to the test, and individuals with active TB are assumed to react with probability one. These states are therefore included when computing the modelled TST positivity used as a calibration target, but no preventive treatment flow originates from them: individuals who have cleared or recovered are not at risk of progression and would derive no benefit from TPT, while those with active disease would be identified and treated through the concurrent disease-screening arm of the same campaign.

Age targeting. Screening programmes are targeted to specific age groups by applying multiplicative coverage adjustments during model age stratification. Age groups not targeted by a given programme have zero screening rate.

Scenario structure. We consider three screening algorithms:

- **PEARL:** the algorithm implemented during the completed phase of the PEARL study. Individuals aged 10 years and above receive CXR, with Xpert additionally used as a frontline test in 35% of them, reflecting operational constraints on its implementation. Children aged 3–9 years receive symptom screening, and TST-guided TPT is offered to all individuals aged 3 years and above.

- **CXR-TST:** Xpert is removed from frontline screening. CXR is used for individuals aged 10 years and above, with symptom screening for ages 3–9 years and TST-guided TPT for ages 3 years and above.

- **Disease screening only:** both Xpert and TST-guided TPT are removed, and screening is restricted to CXR among individuals aged 10 years and above. No screening is performed in children under 10 years, and no preventive treatment is provided.

Each algorithm is combined with each of the three coverage levels, giving a total of nine screening scenarios. These are compared with a baseline scenario in which no active screening is implemented. This framework allows us to assess the incremental population-level contribution of frontline Xpert, of TST-guided preventive treatment, and of increasing screening coverage.

7.3 Treatment outcomes

During treatment, individuals may experience three mutually exclusive outcomes: successful recovery, relapse, or death. Relapse here encompasses all unsuccessful outcomes other than death, including treatment failure and loss to follow-up; these individuals return to the subclinical, less-infectious disease state and remain subject to subsequent detection. Recovery occurs at a rate depending on the time-varying treatment success rate:

$$\text{Recovery:} \quad \text{rate} = \frac{\text{TSR}(t)}{\tau},$$

where $\text{TSR}(t)$ denotes the proportion of successful treatments at time t, and τ is the average treatment duration.

Age-specific relapse and treatment-death rates are calculated to account for both background mortality and the fixed proportion of treatment episodes that result in negative outcomes. Let μ_i be the natural mortality rate for age group i. The fixed fraction of negative outcomes resulting in death is denoted f_{neg}.

The expected proportion of individuals dying from natural causes while on treatment is:

$$p_i^{\text{nat\_death}} = 1 - e^{-\tau\mu_i}.$$

The target proportion of deaths due to treatment failure (excluding natural mortality) is calculated as the fraction of negative outcomes allocated to death:

$$p^{\text{tx\_death}} = (1 - \text{TSR}(t)) \cdot f_{\text{neg}}.$$

To ensure that the overall probability of death on treatment remains non-negative after accounting for natural mortality, the treatment-death rate is defined as:

$$\mu_i^{\text{TX}} = \frac{\max\{p^{\text{tx\_death}} - p_i^{\text{nat\_death}}, 0\}}{\tau}.$$

Finally, the age-specific relapse rate is derived from the residual fraction of unsuccessful treatment episodes, after subtracting both treatment deaths and natural deaths:

$$\phi_i = \frac{1 - \text{TSR}(t) - \tau\mu_i^{\text{TX}} - p_i^{\text{nat\_death}}}{\tau}.$$

These equations ensure that the sum of recovery, relapse, and treatment-death probabilities over the course of treatment equals one, while explicitly incorporating time-varying treatment success and age-specific background mortality.

8 Ordinary Differential Equations

8.1 Notations

We present the ODEs for a generic age group $i \in \{1, \ldots, G\}$ and reachability stratum $s \in \{\text{r}, \text{u}\}$, where r denotes screening-reachable and u hard-to-reach individuals (Section 6). Every compartment therefore carries both indices, e.g. $S_{i,s}$. No flow connects the two reachability strata, so s is a fixed attribute of an individual and the strata are coupled only through the shared force of infection.

Let $\alpha_i(t)$ be the ageing rate from group i to $i{+}1$ ($\alpha_G = 0$), and $\mu_i(t)$ the all-cause mortality rate; neither depends on s. For any compartment variable $U \in \{S, E, C, X, D_{SL}, D_{SH}, D_{CL}, D_{CH}, T, R\}$, define the (within-stratum) ageing operator

$$\mathcal{A}_i[U_{\cdot,s}] \equiv -\alpha_i\, U_{i,s} + \alpha_{i-1}\, U_{i-1,s}, \qquad \alpha_0 = 0.$$

Let $N_{i,s}(t)$ be the total population of age group i in stratum s, and $N_i(t) = \sum_s N_{i,s}(t)$. The force of infection $\lambda_i(t)$ is given above and is common to both strata.

Reachability multipliers. The three multiplicative adjustments that distinguish the two strata (Section 6) are written as stratum-indexed factors:

$$\theta_s^{\text{sus}} = \begin{cases} 1, & s = \text{r}, \\ \theta_{\text{sus}}, & s = \text{u}, \end{cases} \qquad \theta_s^{\text{det}} = \begin{cases} 1, & s = \text{r}, \\ \theta_{\text{det}}, & s = \text{u}, \end{cases} \qquad \theta_s^{\text{scr}} = \begin{cases} 1, & s = \text{r}, \\ 0, & s = \text{u}, \end{cases}$$

applied respectively to all infection and reinfection flows, to the passive detection rate, and to all active screening flows. All other natural-history, treatment and demographic parameters are shared between strata.

Entries into the model. Let $\mathcal{B}(t)$ be the additional entry rate used to reproduce the observed population growth, allocated between strata according to the fixed screening-reachable fraction π, with $\pi_{\text{r}} = \pi$ and $\pi_{\text{u}} = 1 - \pi$. All deaths are recycled into the M.tb-naïve compartment of the youngest age group of the stratum in which they occurred, giving the stratum-specific re-entry term

$$\mathcal{D}_s(t) = \sum_{i=1}^{G} \left[\mu_i N_{i,s} + \sum_{k \in \mathcal{K}} \mu_k^{TB}\, D_{k,i,s} + \mu_i^{TX}\, T_{i,s} \right].$$

Together, these two devices keep the total population on its observed trajectory while holding the screening-reachable and hard-to-reach fractions constant at π and $1 - \pi$.

Early infection parameters (age-specific). For each age group i, let γ_i be the containment rate from E to C, ϵ_i the total progression rate from E to subclinical disease. The proportion progressing initially to "high infectiousness subclinical" rather than "low infectiousness subclinical" is denoted by p. Then progression risks are $\epsilon_i(1-p)$ to D_{SL} and $\epsilon_i p$ to D_{SH}. From C, let ξ be the clearance rate and ω the breakdown rate.

Clinical/infectiousness transitions and recovery. Let χ^+ (subclinical$\to$clinical) and χ^- (clinical$\to$subclinical) be clinical transitions; σ^+ (low$\to$high) and σ^- (high$\to$low) be infectiousness transitions. For notational convenience, we define χ^k to denote χ^+ when k corresponds to a subclinical state and χ^- when k corresponds to a clinical state; analogously, σ^k denotes σ^+ for low infectiousness states and σ^- for high infectiousness states.

Let $\mathcal{K} = \{SL, SH, CL, CH\}$ index the active disease states, where S and C denote subclinical and clinical disease, respectively, and L and H denote low and high infectiousness. Thus, for example, SL represents subclinical disease with low infectiousness. We introduce the notation $\bar{k}$ to denote the state obtained by switching the clinical status of k (e.g., $\overline{SL} = CL$) and $\tilde{k}$ to denote the state obtained by switching the infectiousness status of k (e.g., $\widetilde{SL} = SH$).

Self-recovery from subclinical states occurs at rate ρ_{SC}.

Screening and passive detection hazards. Let $\zeta_{k,i}(t)$ denote the (time- and age-specific) *screening* detection hazard applied to disease state k in the screening-reachable stratum. Screening is unavailable to the hard-to-reach stratum and passive detection is reduced there, while passive detection applies only to *clinical* disease. The total detection hazard is therefore

$$\eta_{k,i,s}^{\text{tot}}(t) = \begin{cases} \theta_s^{\text{scr}}\, \zeta_{k,i}(t), & k \in \{SL, SH\} \quad \text{(subclinical; no passive detection)}, \\ \theta_s^{\text{scr}}\, \zeta_{k,i}(t) + \theta_s^{\text{det}}\, \delta(t), & k \in \{CL, CH\} \quad \text{(clinical; screening + passive)}. \end{cases}$$

Since $\theta_{\text{u}}^{\text{scr}} = 0$, this reduces to $\eta_{k,i,\text{u}}^{\text{tot}} = 0$ for subclinical states and to $\theta_{\text{det}}\, \delta(t)$ for clinical states in the hard-to-reach stratum. The rate of TPT in the screening-reachable stratum is denoted $u_i^{\text{TPT}}(t)$ and combines the TST screening rate and the preventive treatment completion rate; in the hard-to-reach stratum it is set to zero, so that

$$u_{i,s}^{\text{TPT}}(t) = \theta_s^{\text{scr}}\, u_i^{\text{TPT}}(t).$$

State-specific auxiliary terms. The following terms depend on the disease state and, where indicated, on age, but not on the reachability stratum. Subclinical-only self-recovery by state:

$$r_k^{\text{SC}} = \begin{cases} \rho_{SC}, & k \in \{SL, SH\}, \\ 0, & k \in \{CL, CH\}. \end{cases}$$

TB-specific mortality by active state (applied to clinical states only, and depending on infectiousness):

$$\mu_k^{TB} = \begin{cases} 0, & k \in \{SL, SH\}, \\ \mu_{CL}^{TB}, & k = CL, \\ \mu_{CH}^{TB}, & k = CH. \end{cases}$$

Age-specific progression seeding from E_i into active disease:

$$b_{k,i} = \begin{cases} \epsilon_i(1-p), & k = SL, \\ \epsilon_i p, & k = SH, \\ 0, & k \in \{CL, CH\}. \end{cases}$$

8.2 System of ODEs

Recall that i indexes the age group and s the reachability stratum (r: screening-reachable, u: hard-to-reach).

$\forall i \in \{1, \dots, G\}$ and $\forall s \in \{\mathrm{r}, \mathrm{u}\}$:

$$\frac{dS_{i,s}}{dt} = \mathbf{1}_{\{i=1\}} \left(\pi_s \mathcal{B}(t) + \mathcal{D}_s(t)\right) - \theta_s^{\text{sus}} \sigma_i^{\text{child}} \lambda_i S_{i,s} - \mu_i S_{i,s} + \mathcal{A}_i[S_{\cdot,s}], \tag{1}$$

$$\frac{dE_{i,s}}{dt} = \theta_s^{\text{sus}} \lambda_i \left[\sigma_i^{\text{child}} S_{i,s} + \rho_C C_{i,s} + \rho_X (X_{i,s} + R_{i,s})\right] + \omega C_{i,s} - \left(\gamma_i + \epsilon_i + \mu_i + u_{i,s}^{\text{TPT}}\right) E_{i,s} + \mathcal{A}_i[E_{\cdot,s}], \tag{2}$$

$$\frac{dC_{i,s}}{dt} = \gamma_i E_{i,s} - \left(\omega + \xi + \mu_i + u_{i,s}^{\text{TPT}}\right) C_{i,s} + \mathcal{A}_i[C_{\cdot,s}], \tag{3}$$

$$\frac{dX_{i,s}}{dt} = \xi C_{i,s} + u_{i,s}^{\text{TPT}} (E_{i,s} + C_{i,s}) - \theta_s^{\text{sus}} \rho_X \lambda_i X_{i,s} - \mu_i X_{i,s} + \mathcal{A}_i[X_{\cdot,s}], \tag{4}$$

$$\begin{aligned}\frac{dD_{k,i,s}}{dt} = {} & b_{k,i} E_{i,s} + \underbrace{\chi^{\bar{k}} D_{\bar{k},i,s} - \chi^k D_{k,i,s}}_{\text{clinical transitions}} + \underbrace{\sigma^{\tilde{k}} D_{\tilde{k},i,s} - \sigma^k D_{k,i,s}}_{\text{infectiousness transitions}} + \underbrace{\mathbf{1}_{\{k=SL\}} \phi_i T_{i,s}}_{\text{treatment relapse}} \\ & - \left(\eta_{k,i,s}^{\text{tot}}(t) + \mu_i + \mu_k^{TB} + r_k^{\text{SC}}\right) D_{k,i,s} + \mathcal{A}_i[D_{k,\cdot,s}], \qquad k \in \mathcal{K},\end{aligned} \tag{5}$$

$$\frac{dT_{i,s}}{dt} = \sum_{k \in \mathcal{K}} \eta_{k,i,s}^{\text{tot}}(t) D_{k,i,s} - \left(\frac{\text{TSR}(t)}{\tau} + \phi_i + \mu_i^{TX} + \mu_i\right) T_{i,s} + \mathcal{A}_i[T_{\cdot,s}], \tag{6}$$

$$\frac{dR_{i,s}}{dt} = \sum_{k \in \mathcal{K}} r_k^{\text{SC}} D_{k,i,s} + \frac{\text{TSR}(t)}{\tau} T_{i,s} - \theta_s^{\text{sus}} \rho_X \lambda_i R_{i,s} - \mu_i R_{i,s} + \mathcal{A}_i[R_{\cdot,s}]. \tag{7}$$

9 Model Calibration

9.1 Data Sources

Epidemiological observations from PEARL. Measured TB prevalence (using the PEARL algorithm and using CXR), the proportions of prevalent TB that were subclinical and more infectious, and age-specific TST positivity were obtained from the completed phase of PEARL screening. Because these observations were made among screening participants, they are compared with model outputs restricted to the screening-reachable stratum (Section 6). An additional, earlier estimate of TST positivity among adults aged 18 years and over was taken from a previous survey conducted in Kiribati.

TB notifications. Historical TB notifications were obtained from the Kiribati National Tuberculosis Programme, recorded by locality within South Tarawa. Because the model represents the part of South Tarawa remaining to be screened by PEARL in January 2026, the notification series was restricted to the localities east of, and excluding, Betio, Bairiki and Nanikai. A proportion of notifications each year had no locality recorded; these were allocated to the modelled area in proportion to its share of the South Tarawa population (63.3%), giving the imputed annual series used for calibration.

External contact matrix. No empirical contact survey data are available for Kiribati. To constrain the age-mixing matrix estimated within the model (Section 5.2), we generated a synthetic contact matrix using the `conmat` R package. A contact model was fitted to the POLYMOD contact survey data and the corresponding survey populations, and then used to predict all-setting contact rates for the age distribution of the modelled South Tarawa population in 2025, using the same age breaks as the model. The resulting matrix was normalised by its spectral radius so that it is comparable with the model's normalised mixing matrix, and enters the likelihood through the matrix-distance term described below.

9.2 Calibration Targets and Likelihood

Calibration targets include TB notifications, TST positivity by age, prevalence by disease form, fraction subclinical, fraction high infectiousness, and a penalty on the distance between the modelled mixing matrix and the external matrix described in Section 9.1. For prevalence targets, model outputs incorporate test sensitivity parameters. Each target k consists of one or more observations $y_{k,t}$ made at times t, each compared with the corresponding model output $m_{k,t}(\theta)$ through a normal likelihood:

$$y_{k,t} \sim \mathcal{N}\big(m_{k,t}(\theta), \sigma_k^2\big).$$

A single standard deviation is used for all observations belonging to a given target, defined relative to the mean of the observed series $\bar{y}_k$:

$$\sigma_k = \frac{\mathrm{tol}_k}{1.96}\,\bar{y}_k, \qquad \mathrm{tol}_k = \begin{cases} 0.40, & k = \text{notifications}, \\ 0.20, & \text{otherwise}, \end{cases}$$

so that the 95% interval of the likelihood spans $\pm 20\%$ (or $\pm 40\%$ for notifications) of the mean observed value. The total log-likelihood is

$$\log L(\theta) = \sum_k \sum_t \Big(- \frac{(y_{k,t} - m_{k,t}(\theta))^2}{2\sigma_k^2} - \frac{1}{2}\log(2\pi\sigma_k^2)\Big).$$

Mixing matrix penalty. The estimated mixing matrix is not compared with the external synthetic matrix element by element. Instead, the discrepancy between the two is summarised by a single scalar, the Canberra distance between the spectral-radius-normalised model matrix $\tilde{\mathbf{C}}(t)$ evaluated in 2025 and the normalised external matrix $\mathbf{C}^{\mathrm{ext}}$:

$$d(\theta) = \sum_{i,j} \frac{\big|\tilde{C}_{ij} - C_{ij}^{\mathrm{ext}}\big|}{\big|\tilde{C}_{ij}\big| + \big|C_{ij}^{\mathrm{ext}}\big| + \varepsilon},$$

with $\varepsilon = 10^{-10}$ included to avoid division by zero. This relative, element-wise normalisation gives comparable weight to discrepancies in low- and high-contact cells, rather than allowing the largest matrix entries to dominate.

The distance is treated as an additional target with an observed value of zero, $0 \sim \mathcal{N}\big(d(\theta), \sigma_{\mathrm{mix}}^2\big)$, so that the likelihood penalises mixing patterns that depart from the external matrix. Unlike the epidemiological targets, the standard deviation σ_{mix} is not fixed but estimated during calibration under a uniform prior on $[5, 20]$. This avoids having to specify *a priori* how closely the model matrix should reproduce a synthetic matrix of unknown accuracy for this setting: the strength of the penalty is instead determined by the extent to which the epidemiological data and the external matrix can be reconciled. The three mixing parameters (background level, assortativity spread, and parent–child strength) are therefore informed jointly by this penalty and by the epidemiological targets.

9.3 Metropolis Algorithm

We perform calibration using Markov chain Monte Carlo (MCMC) sampling with the `DEMetropolisZ` algorithm implemented in `PyMC`. We run eight parallel chains to improve exploration of the posterior distribution. Each chain uses 10,000 tuning (adaptation) iterations, followed by 20,000 posterior draws, giving 240,000 model evaluations per calibration. For the subsequent scenario projections, we discard an initial burn-in period of 10,000 draws and retain 2,000 posterior samples per calibrated run.

Convergence is assessed by inspecting trace plots, posterior distribution smoothness, and the Gelman-Rubin diagnostic ($\hat{R}$) being below 1.05. Posterior samples from the calibrated model are propagated forward to generate projections under each intervention scenario, thereby fully accounting for parameter uncertainty.

Each model configuration is calibrated independently, and the configurations are run in parallel as array jobs on a high-performance computing cluster, each using eight cores. The full analysis required between approximately 10 and 15 hours, depending on the parameter values explored.

10 Parameter Definitions, Values and Priors

In this section, we describe the sources and rationale underlying the parameter values and prior distributions used in the model. The complete list of parameter definitions, numerical values, calibration priors, and references is provided in Table 1, with the sensitivities of the screening approaches reported separately in Table 2. Parameters were informed using a combination of published literature, previous modelling studies, empirical observations from Kiribati, and calibration to epidemiological data. Where direct evidence was unavailable or substantial uncertainty existed, prior distributions were specified to reflect plausible ranges and updated through Bayesian calibration.

Unless stated otherwise, priors on estimated parameters are uniform and were deliberately specified over ranges wide enough not to constrain the posterior, so that the calibration targets are the only quantities informing their values. Where a bound is not derived from published evidence, it was instead chosen so that the values it excludes correspond to model behaviour that is either mechanistically impossible or plainly incompatible with the observed data.

Table 1 summarises the parameter definitions, values, and priors used in the model, while the following sections provide detailed descriptions and justifications for the parameter choices.

Table 1: Model parameters. $\mathcal{U}(a, b)$ denotes a uniform prior on the interval $[a, b]$ for parameters estimated during calibration; all other entries are fixed values. The Source column gives the origin of the value or of the prior bounds, and the rationale for each is set out in Section 10.

Parameter	Value / Prior	Source
Transmission and mixing		
Effective rate of transmission (before adjusting for infectiousness)	$\mathcal{U}(0.0001, 0.003)$	Calibrated
Population-size exponent in the force of infection, from density-dependent (0) to frequency-dependent (1) transmission	$\mathcal{U}(0, 1)$	Full range of possible values
Background age-agnostic mixing level	$\mathcal{U}(0.01, 0.05)$	Early model exploration
Spread of assortative mixing pattern (smaller value means more assortativity)	$\mathcal{U}(2, 15)$	Early model exploration
Strength of parent-children mixing pattern	$\mathcal{U}(0.01, 1)$	Early model exploration
Rel. susceptibility to infection for under 15 year-old individuals (ref. 15 and over)	$\mathcal{U}(0.3, 1)$	[1–3]
Infection and early progression		
Rate of progression from 'incipient' to TB disease (age 0-4, /year)	2.4	[4]
Rate of progression from 'incipient' to TB disease (age 5-14, /year)	2	[4]
Rate of progression from 'incipient' to TB disease (age 15 and over, /year)	0.1	[4]
Rate of transition from 'incipient' to 'contained' (age 0-4, /year)	4.4	[4]
Rate of transition from 'incipient' to 'contained' (age 5-14, /year)	4.4	[4]
Rate of transition from 'incipient' to 'contained' (age 15 and over, /year)	2	[4]
Rate of transition from 'contained' to 'incipient' (all ages, /year)	$\mathcal{U}(0.01, 1)$	Calibrated
Rate of transition from 'contained' to 'cleared' (all ages, /year)	$\mathcal{U}(0.01, 0.1)$	[5, 6]
Rel. risk of reinfection from 'contained' compartment (ref. 'mtb-naïve')	$\mathcal{U}(0.2, 0.5)$	[7]
Rel. risk of reinfection from 'cleared' and 'recovered' compartment (ref. 'mtb-naïve')	$\mathcal{U}(0.5, 1)$	[8, 9]
Active TB disease		
Rate of progression from subclinical to clinical TB (/year)	$\mathcal{U}(0.5, 5)$	Calibrated
Rate of regression from clinical to subclinical TB (/year)	1	Assumption, varied in sensitivity analysis
Rate of progression from 'less infectious' to 'more infectious' TB (/year)	$\mathcal{U}(0.5, 10)$	Calibrated
Rate of regression from 'more infectious' to 'less infectious' TB (/year)	1	Assumption, varied in sensitivity analysis
Rel. infectiousness of subclinical TB (ref. clinical TB)	0.5	Assumption
Rel. infectiousness of 'less infectious' TB (ref. 'more infectious' TB)	0.4	[10–12]

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Table 1 continued from previous page

Parameter	Value / Prior	Source
Rate of TB mortality for 'more infectious' clinical disease (/year)	0.389	[13]
Rate of TB mortality for 'less infectious' clinical disease (/year)	0.025	[13]
Rate of self-recovery for subclinical TB (/year)	0.4	[13]
Passive detection and treatment		
Annual rate of TB detection in 2020	$\mathcal{U}(0.1, 5)$	Calibrated
Inflection year of the passive detection scale-up profile	$\mathcal{U}(1990, 2020)$	Calibrated
Shape parameter of passive detection scale-up profile	0.15	Early model exploration
Past passive detection rate, as a fraction of the current one	$\mathcal{U}(0.3, 1)$	Calibrated
Average TB treatment duration (years)	0.5	Standard six-month regimen
Percentage of deaths among non-successful TB treatment outcomes	40	Treatment outcomes reported to WHO for Kiribati
TPT completion percentage	70	Observed during PEARL, varied in sensitivity analysis
Screening reachability		
Proportion of population that is reachable by screening interventions	0.85	Observed during PEARL
Relative detection rate of hard-to-reach population under passive case finding (ref. screening-reachable)	0.5	Assumption
Relative susceptibility to infection of hard-to-reach population (ref. screening-reachable)	1.5	Assumption, varied in sensitivity analysis

10.1 Demographic inputs

Population size by single year of age (1950–2100), age-specific deaths, and age-specific fertility rates (1950–2023) were taken from the United Nations World Population Prospects. These series are data inputs rather than estimated parameters, and the way they enter the model is described in Sections 3 and 5.2.

10.2 TB natural history parameters

The model incorporates a detailed representation of the spectrum of *M. tuberculosis* infection and disease. The parameters governing this representation are described below in the order in which an individual encounters them: progression and containment following infection, breakdown and clearance of established infection, susceptibility to reinfection, transitions across the active disease spectrum, relative infectiousness, and the outcomes of untreated disease.

10.2.1 Progression and containment following infection

In the present model, newly infected individuals enter an incipient infection state (E), from which they may either progress to active disease or transition to a contained infection state (C). Contained infection may subsequently undergo immune-mediated clearance to X or return to the incipient state following immunological breakdown.

All rates governing progression out of the incipient infection state were informed by our previous work on TB latency parameterisation, which estimated parameters for different model structures using TB progression data [4]. While the model structure used in this analysis differs from those previously calibrated, we can establish a correspondence between key parameters based on their role within the early infection process.

The previous parameterisation study calibrated several alternative latent infection structures. The comparison presented here focuses on Model 4, which represented early infection using two latent compartments. Individuals entered an initial latent state and could either progress directly to active TB or transition to a secondary latent state associated with slower progression dynamics. The parameter ϵ governed direct progression from the initial latent state to disease, while the parameter κ governed transition from the initial latent state into the secondary latent state.

The progression parameter in the present model, denoted ϵ_i, governs direct progression from the initially occupied infection state (E) to active disease. This parameter therefore has the same interpretation as ϵ in Model 4, namely the rate at which newly infected individuals progress directly to disease before entering a more controlled infection state. This correspondence can be demonstrated analytically by considering disease incidence immediately following infection. When all individuals are initialised in the high-risk infection state, the incidence rate at $t = 0$ reduces to the direct progression parameter in both model structures, establishing the equivalence of ϵ and ϵ_i with respect to early progression dynamics. Consequently, estimates of ϵ from the previous calibration were used directly to inform age-specific values of ϵ_i.

No direct mathematical correspondence exists between the containment rate in the present model (γ_i) and the transition rate κ estimated in Model 4. The present formulation includes additional pathways, notably immune breakdown from the contained state back to the incipient state, which were not represented in the earlier model. Nevertheless, κ provides the most relevant source of information because both parameters govern movement from an initially infected state with elevated risk of disease progression into a state associated with greater immune control and lower short-term progression risk. Estimates of κ were therefore used to inform γ_i in the present model.

10.2.2 Immune breakdown and clearance of infection

Importantly, the revised model allows the balance between rapid progression and longer-term latent infection to be determined jointly by the containment rate (γ_i) and the immune breakdown rate (ω). Calibration of the breakdown parameter therefore provides flexibility to reproduce a range of early infection trajectories that could not be represented within the simpler Model 4 structure.

No direct estimate of ω is available, since loss of immune control is not directly observable, and the prior was therefore chosen to be wide enough not to constrain the posterior. Its bounds of 0.01 and 1.0 per year correspond to an average time spent in the contained state before breakdown of approximately one hundred years at one extreme and one year at the other. The lower bound therefore represents contained infection that is effectively lifelong for the great majority of individuals, and the upper bound a state that provides only transient control, so that the interval brackets the full range of behaviours the compartment could plausibly represent. The posterior is consequently determined by the calibration targets rather than by the prior.

The remaining parameter governing early infection dynamics, the clearance rate (ξ), describes spontaneous resolution of contained infection and has no direct analogue in the previous model. Infection with *M. tuberculosis* is increasingly recognised not to be lifelong, with a substantial proportion of infected individuals eventually clearing the organism [6]. A Bayesian analysis combining post-mortem and tuberculin skin test reversion data estimated that around a quarter of infected individuals self-clear within ten years of infection, and roughly three quarters over a lifetime [5]. The prior for ξ was specified to span this range of behaviours and was updated through calibration.

10.2.3 Susceptibility to reinfection

Susceptibility to reinfection is governed by two parameters. The first applies to individuals in the contained infection state, and the second to individuals who have cleared infection or recovered from disease. Both were assigned uniform priors reflecting current uncertainty in the literature and updated through calibration, and both were restricted to values not exceeding one, so that neither state is more susceptible than a previously uninfected individual.

For the contained state, the prior was centred on the protection implied by a meta-analysis of pre-chemotherapy cohort studies, which estimated an incidence rate ratio for tuberculosis following reinfection of around 0.2 in individuals with latent infection relative to uninfected individuals, corresponding to a reduction in risk of roughly 80% [7]. For individuals who have cleared infection or recovered from disease, the evidence points to considerably weaker protection: in a high-incidence setting the age-adjusted incidence of tuberculosis attributable to reinfection after successful treatment was approximately four times the rate of new disease [8], and exogenous reinfection accounts for an appreciable share of late recurrence even in low-incidence settings [9].

These observations should not be read as direct estimates of biological susceptibility. Individuals who have previously had tuberculosis are not a random sample of the population: having developed disease once identifies people who live in the settings of highest transmission intensity and who carry the socio-economic disadvantage, crowding, undernutrition and comorbidities that predisposed them to disease in the first place, and these characteristics persist after treatment. The elevated rate of recurrent disease in this group therefore reflects a combination of continued high exposure, persistent host risk factors, and any residual lung damage, rather than an intrinsic loss of protection conferred by the earlier infection; the model represents exposure through the force of infection but does not stratify by these individual-level risk factors, so the parameter cannot absorb their effect. The prior for this second parameter accordingly spans weak-to-absent protection, extending up to one, the structural upper limit imposed in the model, but is not permitted to exceed it.

10.2.4 Transitions across the active disease spectrum

The active disease component distinguishes individuals according to both clinical status (subclinical versus clinical disease) and infectiousness (less infectious versus more infectious disease). Parameters governing transitions between these disease states were selected to reproduce realistic distributions of disease phenotypes while remaining identifiable from the available data. To reduce identifiability issues, transition rates from more infectious to less infectious disease and from clinical to subclinical disease were fixed at values corresponding to an average duration of approximately one year in each state before transition. Transition rates in the opposite direction were estimated during calibration.

The priors for these two estimated rates (χ^+ for subclinical $\rightarrow$ clinical progression and σ^+ for gain of infectiousness) are best interpreted relative to the corresponding regression rates, which are fixed at 1.0 per year, since it is the balance between the two directions that determines the distribution of prevalent TB across states. The bounds were deliberately set wide, spanning distributions ranging from a large predominance of subclinical and less-infectious disease to the opposite, so that the posteriors are determined by the distribution of disease states measured during the completed PEARL screening phase rather than by the prior.

10.2.5 Relative infectiousness

Relative infectiousness parameters were informed by the literature on heterogeneity in transmission potential where relevant estimates were available, and set by assumption where they were not.

Individuals classified as less infectious were assumed to be 40% as infectious (per unit time) as those classified as more infectious. Molecular-epidemiological studies that attribute secondary cases to putative source cases have estimated the transmission rate of smear-negative relative to smear-positive source cases at around 20% [10–12]. The value used here is deliberately higher, because the model's "less infectious" category is defined by the classification available from PEARL rather than by smear status alone, and therefore includes some disease that would be classified as smear-positive.

Subclinical disease was likewise assumed to be half as infectious as clinical disease per unit time. We are not aware of a definitive estimate of this quantity. The value is necessarily below one, since symptoms such as cough are known to promote the generation of infectious aerosol. It is unlikely to be very much below one, however: a considerable proportion of the subclinical disease detected in community surveys is sputum-smear positive, suggesting appreciable infectiousness [14], and cough is in any case not a prerequisite for transmission [15]. The value adopted is therefore a modelling assumption rather than an estimate. The combined infectiousness of each disease category was determined by multiplying the corresponding relative infectiousness factors.

10.2.6 Outcomes of untreated disease

Rates of TB mortality and of self-recovery were taken from a Bayesian re-analysis of pre-chemotherapy cohort data, which estimated the natural history parameters of untreated pulmonary tuberculosis [13], and held fixed; the states to which each applies are set out in Section 4. Relapse following unsuccessful treatment is not a free parameter, but is derived from the treatment success rate, the fraction of unsuccessful outcomes ending in death, and age-specific background mortality, as described in Section 7. These quantities were fixed rather than estimated because they are only weakly identifiable from the cross-sectional data available, and because they were not expected to materially affect the comparison between screening algorithms that is the focus of this analysis; estimating them would have introduced identifiability problems without a corresponding gain.

10.3 Transmission, mixing and BCG protection

10.3.1 Population-size scaling of transmission

The model includes a transmission-scaling parameter that allows a continuum between frequency-dependent and density-dependent transmission. This parameter was introduced to account for uncertainty regarding the relationship between population density and transmission intensity in Kiribati, where increasing population density and limited habitable land may plausibly influence contact patterns and transmission risk. The parameter was assigned a uniform prior spanning the full range between purely frequency-dependent and purely density-dependent transmission.

10.3.2 Age-mixing parameters

No contact survey has been conducted in Kiribati, so the age-mixing matrix is built from three estimated parameters rather than taken directly from data: a uniform background contact level, the spread of the assortative kernel, and the strength of parent–child contacts (Section 5.2). These parameters were varied during calibration in order to account for uncertainty in age-mixing patterns. The prior ranges were established by generating the mixing matrices implied by candidate values and inspecting them visually, retaining the range over which the resulting matrices remained plausible. Outside these bounds the matrices became either almost entirely assortative, with negligible contact between age groups, or effectively homogeneous, with the age structure of contacts lost. Within these bounds the three parameters were expected to be informed jointly by the age-specific TST positivity data and by the penalty on the distance between the modelled matrix and an external synthetic matrix, described in Section 9.1.

10.3.3 BCG protection in children

Infection flows in M.tb-naive individuals aged under 15 years are multiplied by a factor representing the protection conferred by BCG vaccination, which is administered at birth in Kiribati. BCG is generally accepted to reduce the risk of acquiring infection, in addition to reducing the risk of progression once infected, but estimates of the magnitude and the duration of this protection vary substantially between settings and study designs [1–3]. A meta-analysis of children with known recent exposure estimated protection against infection of around 20% in vaccinated compared with unvaccinated children, rising to roughly a quarter in the subset of studies with information on progression to disease at the time of screening [1]. The parameter was therefore given a deliberately wide uniform prior, spanning a 70% reduction in susceptibility at one extreme and no protection at all at the other, and was estimated during calibration.

10.4 Programmatic parameters

The structural representation of passive detection, treatment and screening is described in Section 7, and that of screening reachability in Section 6. This section documents the values assigned to the associated parameters, the evidence used to inform them, and the reasons for estimating some during calibration while fixing others.

10.4.1 Passive detection

No direct estimate of the passive detection rate was available for Kiribati. The recent detection level (δ_{rec}), the inflection time of the scale-up (t^*) and the past-to-recent ratio (ϕ) were therefore assigned uninformative uniform priors and estimated during calibration against the historical notification series (Table 1). The prior on δ_{rec} spans a fifty-fold range. In the absence of competing transitions, its bounds would correspond to an average interval between the onset of clinical disease and treatment initiation of between roughly two months and ten years. The interval actually implied by the model is shorter than this, because individuals with clinical disease may instead regress to subclinical disease, die, or age into another stratum before being detected. The range was deliberately made wide so that the posterior is determined by the notification data rather than by the prior.

The steepness parameter σ was fixed at 0.15 per year rather than estimated, since the timing and the abruptness of the scale-up are only weakly separable given a single notification time series. The value was chosen by plotting the detection profiles implied by candidate values and retaining one that produced a gradual increase spread over several decades, consistent with the progressive strengthening of diagnostic capacity and access to care described in Section 7. An example profile generated with this value is shown in Figure 2.

The relative detection rate for subclinical disease was fixed at zero and the relative detection rate in the hard-to-reach stratum at 0.5, following the assumptions set out in Sections 7 and 6. Neither quantity is identifiable from the available data.

10.4.2 Treatment

The average treatment duration τ was fixed at 0.5 years, corresponding to the standard six-month first-line regimen.

The treatment success rate $\text{TSR}(t)$ is a time-variant data input rather than an estimated parameter. Annual reported values for Kiribati were used for 1995–2022, together with a recent value for 2025 and two earlier anchor points representing the era before effective chemotherapy (0% in 1945) and its early introduction (50% in 1950). Values are linearly interpolated between these points and held constant outside the range. The treatment success rate determines the recovery rate on treatment and, together with the fixed fraction of unsuccessful episodes ending in death, the relapse and treatment-death rates described in Section 7.

The fraction of negative outcomes resulting in death, f_{neg}, was fixed at 40%, approximating the proportion of unsuccessful outcomes recorded as deaths in the treatment outcomes reported to WHO for Kiribati. The remaining unsuccessful outcomes, comprising treatment failure, loss to follow-up and cases not evaluated, are returned to active disease in the model. This proportion fluctuates markedly between years because the annual cohorts are small, so a single representative value was used rather than a time-varying series, and it was not estimated during calibration.

10.4.3 Reachability parameters

The screening-reachable fraction π was fixed at 0.85, based on participation observed during the completed phase of the PEARL study. The relative passive detection rate in the hard-to-reach stratum (θ_{det}) was fixed at 0.5. No epidemiological observation distinguishes the two strata, so this quantity is not identifiable and could not be estimated.

The relative susceptibility of the hard-to-reach stratum (θ_{sus}) is equally unidentifiable, but it has a far greater bearing on the projected impact of screening, because it governs how much transmission persists in the group that screening cannot reach. This parameter was fixed at 1.5 in the base-case configuration, and varied across 1.0, 1.5, 2.0 and 3.0 in sensitivity analyses.

10.4.4 Screening algorithm sensitivities

The sensitivity assigned to each screening algorithm for each model state is reported in Table 2.

With the exception of the TST parameters, these values were derived from the completed phase of the PEARL campaign, in which each participant found to have TB was classified by clinical status and

Table 2: Sensitivity of each screening approach to the model states it can detect, i.e. the probability that an individual occupying a given state and screened under that approach is correctly identified. $\mathcal{U}(a, b)$ denotes a uniform prior estimated during calibration; dashes indicate states that the approach does not target. Because operational constraints meant that frontline Xpert could be applied to only 35% of those screened, the PEARL algorithm is represented as the corresponding weighted mixture of its two branches, $s_k^{\text{PEARL}} = 0.35\, s_k^{\text{Xpert+CXR}} + 0.65\, s_k^{\text{CXR}}$.

Screening approach	Incipient infection	Contained infection	Cleared or recovered	Subclinical low inf.	Clinical low inf.	Subclinical high inf.	Clinical high inf.
Tuberculin skin test (TST)	0.75	0.75	$\mathcal{U}(0.2, 0.5)$	–*	–*	–*	–*
Symptom screening (SSx)	–	–	–	0	1	0	1
Chest X-ray (CXR)	–	–	–	$\mathcal{U}(0.07, 0.52)$	1	$\mathcal{U}(0.72, 0.93)$	1
Xpert + CXR	–	–	–	1	1	1	1

*These states are not targeted by TST-guided preventive treatment, so no screening flow originates from them. They are nevertheless assumed to be TST-positive with probability one when computing the modelled TST positivity used as a calibration target.

by infectiousness, and the result of each nested algorithm was recorded. Sensitivity was evaluated against all disease detected by the full symptom-screening, CXR and Xpert algorithm among participants with an interpretable Xpert result. That algorithm therefore has a sensitivity of one for every disease state by construction, because it defines the reference standard, and this is the value assigned to the Xpert-containing branch in the model. It should be noted that this branch is not the same as the delivered PEARL programme, which combines it with CXR alone in the proportions given in Section 7.

The sensitivity of CXR to subclinical disease was the only screening quantity that was materially uncertain, and it was estimated during calibration. The prior bounds were derived from the exact (Clopper–Pearson) interval around the sensitivity observed in each subclinical stratum during the completed PEARL phase: 4 of 16 individuals with less-infectious subclinical disease were identified, giving a uniform prior on $[0.07, 0.52]$, against 44 of 52 with more-infectious subclinical disease, giving a uniform prior on $[0.72, 0.93]$. The considerably wider prior for the less-infectious state reflects the small number of individuals in that stratum rather than any additional biological uncertainty. These two parameters are further informed during calibration by the CXR-measured prevalence target, which is compared with a model output constructed using the same sensitivities, alongside the prevalence measured under the full algorithm.

The probability of a positive TST was fixed at 0.75 for both the incipient and contained infection states. The TST positivity data used for calibration were collected using a 10 mm induration threshold, and the model parameter is defined accordingly. No reference standard exists for infection, so the only directly measurable analogue is the sensitivity of the test against bacteriologically confirmed active tuberculosis, which the most recent systematic review pooled at above 80% at this threshold [16]. A slightly lower value was adopted here because the parameter applies to infection rather than disease, and tuberculin reactivity is expected to be weaker in the absence of the higher bacillary burden and stronger antigenic stimulation associated with active disease.

The corresponding probability among individuals who have cleared infection or recovered from disease was estimated during calibration under a uniform prior on $[0.2, 0.5]$, since the persistence of tuberculin reactivity after clearance is poorly characterised. The bounds were informed by the frequency with which infection tests revert to negative: a meta-analysis of serial QuantiFERON testing reported a pooled reversion proportion of approximately 25% [17], reversion is common following preventive treatment [18], and reversion rates in a high-burden setting were lower than those reported from low-burden settings [19]. The prior therefore allows between a fifth and a half of individuals who have cleared or recovered to remain test-positive.

The probability of completing preventive treatment once identified as TST-positive was fixed at 0.7, matching the completion proportion observed among participants offered TPT during the completed phase of the PEARL campaign.

10.4.5 Screening coverage

Coverage levels of 65%, 75% and 85% of the total population were examined. These are scenario assumptions rather than estimated quantities: the lower two represent plausible degrees of incomplete participation, while the highest corresponds to essentially complete coverage of the screening-reachable stratum and is treated as the practical upper bound achievable without extending the campaign to the hard-to-reach population (Section 6).

10.5 Sensitivity analyses

In addition to the sixteen configurations that resulted from varying the relative susceptibility of the hard-to-reach stratum and the common disease-regression rate, three further analyses were run in which a single assumption was altered and the model then fully recalibrated.

The first assumed a preventive treatment completion probability of 0.6 rather than the 0.7 observed during the completed PEARL phase, since completion is a key determinant of the benefit attributable to the preventive treatment component of the campaign and may be lower in the remaining area.

The second replaced the calibration target for the proportion of prevalent disease that is subclinical, reducing it from the 81.6% measured during the completed PEARL screening phase to 50%. This addresses the possibility that the measured proportion overstates the true subclinical fraction, and tests how strongly the projected benefit of radiographic screening depends upon it.

The third imposed homogeneous mixing by age. The three mixing parameters and the contact-matrix penalty were removed from the calibration and the model was refitted with all age groups contacting one another at equal rates, isolating the contribution of age-assortative mixing to the projected impact of age-targeted screening.

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